PRODUCT STRATEGY + TECHNICAL BLUEPRINT

School Learning Operations Platform

A multi-tenant SaaS for school administration, teacher workflows, student progress and learning evidence.

INITIAL WEDGE	ARCHITECTURE	MARKET ENTRY
Daily learning loop	Modular monolith	India private schools

Version 1.0 | 6 September 2026

The company is not won by having the most features. It is won by getting schools live quickly, making teachers use the product every week, protecting student data, and proving renewal value.

Prepared as an India-first strategy hypothesis. Pricing, market segment and roadmap must be validated with paid design-partner schools.

01 | PRODUCT THESIS

Executive summary

The proposed business is a multi-tenant B2B SaaS sold to schools. The platform company operates a control plane; each school operates an isolated tenant; teachers run classroom workflows; students consume learning activities and see progress. The product is broader than a traditional LMS, but it should enter the market through a narrow learning-operations wedge rather than a complete school ERP.

Recommended positioning: Bring attendance, assignments, recordings, practice and progress into one student learning timeline - live for a school within seven days.

Decision	Recommendation	Reason
Initial customer	Private CBSE/ICSE schools, 500-3,000 students, one region	Narrow sales motion and repeatable onboarding
Initial product	Daily classroom loop + student learning timeline	Solves a frequent workflow with visible value
Architecture	Next.js + NestJS modular monolith + PostgreSQL	Fast delivery without distributed-system overhead
Data	PostgreSQL only; JSONB for flexible payloads	MongoDB adds cost and consistency problems
Jobs/cache	PostgreSQL-backed jobs first; Redis when measured	Avoid infrastructure before demand
AI	OpenAI from the NestJS worker; teacher approval required	No Python service until Python-specific work exists
Commercial model	Annual contract + onboarding fee + metered expensive usage	Protects cash flow and support economics

What must be true before scaling

- A school can go live in seven days without engineering work.
- Most invited teachers become weekly active users.
- Schools renew because the platform improves visibility and reduces manual work.
- Tenant isolation, auditability and data export are trustworthy.
- Customer acquisition and onboarding costs can be recovered within the first contract year.

02 | WHAT THE PRODUCT SOLVES

Problem, users and value

User	Current pain	Product outcome
School owner / principal	Data is fragmented across spreadsheets, chat and disconnected tools	One view of adoption, attendance, work completion and intervention needs
School administrator	Roster setup, academic changes and reporting require repeated manual work	Imports, reusable configuration, reports and controlled tenant administration
Teacher	Attendance, assignment distribution, grading and reminders consume teaching time	A fast class workspace with reusable content and assisted feedback
Student	Work, recordings, feedback and deadlines are spread across channels	A single timeline showing what happened, what is due and what improved
Parent / guardian	Progress is discovered late or only during formal meetings	A concise digest and later a controlled family portal
Platform operator	Every school requests different workflows and support	Entitlements and constrained configuration without tenant-specific code

The core learning loop

Step	Action	Evidence created
1. Prepare	School imports users and classes; teacher prepares a session	Roster, schedule and class ownership
2. Teach	Teacher records attendance and shares materials or recording	Session, attendance and resources
3. Assess	Teacher posts an assignment or Arena practice set	Expected work, due date and rubric
4. Respond	Student submits, practises and receives immediate objective feedback	Attempts, answers and completion
5. Improve	Teacher grades and school reviews patterns	Feedback, scores, mastery and interventions

North-star metric: weekly classes completing a learning loop - activity published, student work recorded, and teacher feedback returned.

The product must optimize teacher time first. School dashboards are valuable only when teachers consistently create reliable classroom data.

03 | INDIA-FIRST MARKET ENTRY

Market opportunity and competition

1.47M	339,583	95.9M
Total schools in India, UDISE+ 2024-25	Private unaided recognized schools	Students in private unaided schools

These numbers describe the broad market, not the first obtainable market. The initial serviceable market should be one region, one school profile and a limited academic model. A practical first segment is private schools with 500-3,000 students that still coordinate learning through spreadsheets, messages and separate tools.

Competitor	Strength	Opening for this product
Google Classroom	Familiar, easy assignment distribution, Google ecosystem	School-wide operations, cross-class progress and guided onboarding
Canvas / Schoology	Mature LMS, assessment, analytics and integration ecosystems	Localized workflow, faster implementation and India-first service
Moodle	Flexible, extensible and established; Workplace supports multi-tenancy	Lower operating burden, opinionated workflows and accountable support
Teachmint	India brand, learning + school platform breadth, classroom hardware	Focused software experience and measurable teacher/student adoption
Fedena / school ERPs	Broad administration, fees, payroll, transport and records	Deep daily learning loop instead of ERP feature breadth

Competitive conclusion

Feature parity is not a winning strategy. Existing products already offer large feature catalogs. The initial advantage must be implementation speed, teacher usability, local onboarding, transparent progress and trustworthy tenant isolation. AI is useful, but it is not a moat because competitors can access similar models.

04 | FOCUS BEFORE BREADTH

Initial direction and product scope

Target promise

A school can move from roster import to active classroom workflows in seven days, and leadership can see where learning activity is happening without chasing spreadsheets.

Build for the first paid release	Defer until customer evidence
Tenant setup, branding and fixed roles	Custom role builder and arbitrary permissions
Academic year, terms, subjects and classes	Complex multi-campus structures
CSV roster import and export	Deep SIS integrations and certification
Attendance, assignments, submissions and gradebook	Admissions, payroll, transport, hostel and inventory
Student timeline, progress and basic reports	Separate analytics warehouse and predictive risk models
Arena objective practice and teacher question bank	Adaptive learning engine and public leaderboards
Email and in-app notifications	Native mobile apps and broad messaging automation
Files plus external video provider	Building video hosting or transcoding
Teacher-reviewed AI assistance	Autonomous grading or student decisions

Arena MVP

- Question bank tagged by subject, chapter, topic and difficulty.
- MCQ, multi-select, true/false and numeric answers first.
- Practice sets, attempt history, explanations and topic mastery.
- Teacher dashboard for participation and weak-topic patterns.
- AI may draft questions and explanations, but a teacher approves publication.

Avoid public rankings by default. Use personal improvement, streaks and optional class challenges. Competition involving minors requires careful school policy and privacy choices.

05 | ROLES, LEARNING LOOP AND DECISIONS

Full product operating flow

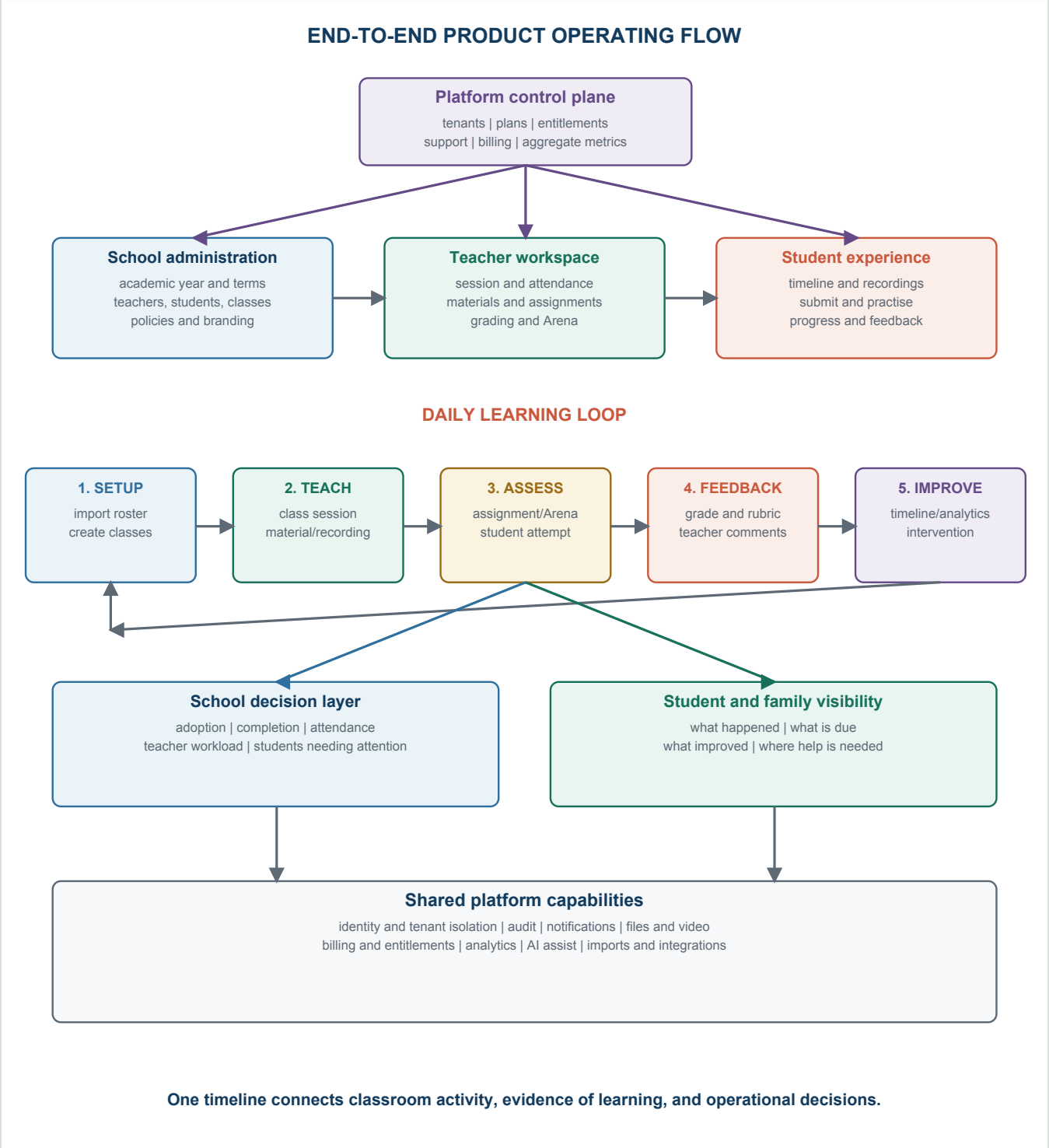


Figure 1. Platform control, tenant operations and the learning loop converge on a shared evidence layer. The school manages its own users and academic operations; the platform manages tenancy, plans, support and service health.

06 | WHAT MANAGES WHAT

Product modules and ownership

Module	Primary owner	Responsibilities
Platform	Internal super admin	Tenant lifecycle, plan, entitlements, support and aggregate usage
Identity and tenancy	Platform + school admin	Users, memberships, authentication, roles and tenant context
Academics	School admin	Academic year, terms, subjects, classes, teachers and enrollment
Classroom	Teacher	Sessions, timeline entries, materials, announcements and recordings
Attendance	Teacher / admin	Session attendance, correction history, imports and reports
Assignments and grading	Teacher	Tasks, submissions, rubrics, grades, feedback and scorecards
Arena	Teacher + student	Question bank, practice sets, attempts, explanations and mastery
Notifications	System	In-app/email delivery, preferences, retries and delivery status
Billing	Platform	Contracts, plan state, invoice references and payment-provider events
Analytics	School + platform	Tenant metrics, teacher adoption and permitted aggregate platform metrics
AI	Teacher-supervised	Draft questions, rubrics, feedback and summaries with cost controls

Platform access boundary

The platform super admin controls tenant lifecycle, entitlements, billing and service support. It must not silently inspect student records or modify grades. A support session should be tenant-approved, time-limited, reason-coded and written to an immutable audit trail.

07 | SIMPLE NOW, SEPARABLE LATER

Technical architecture

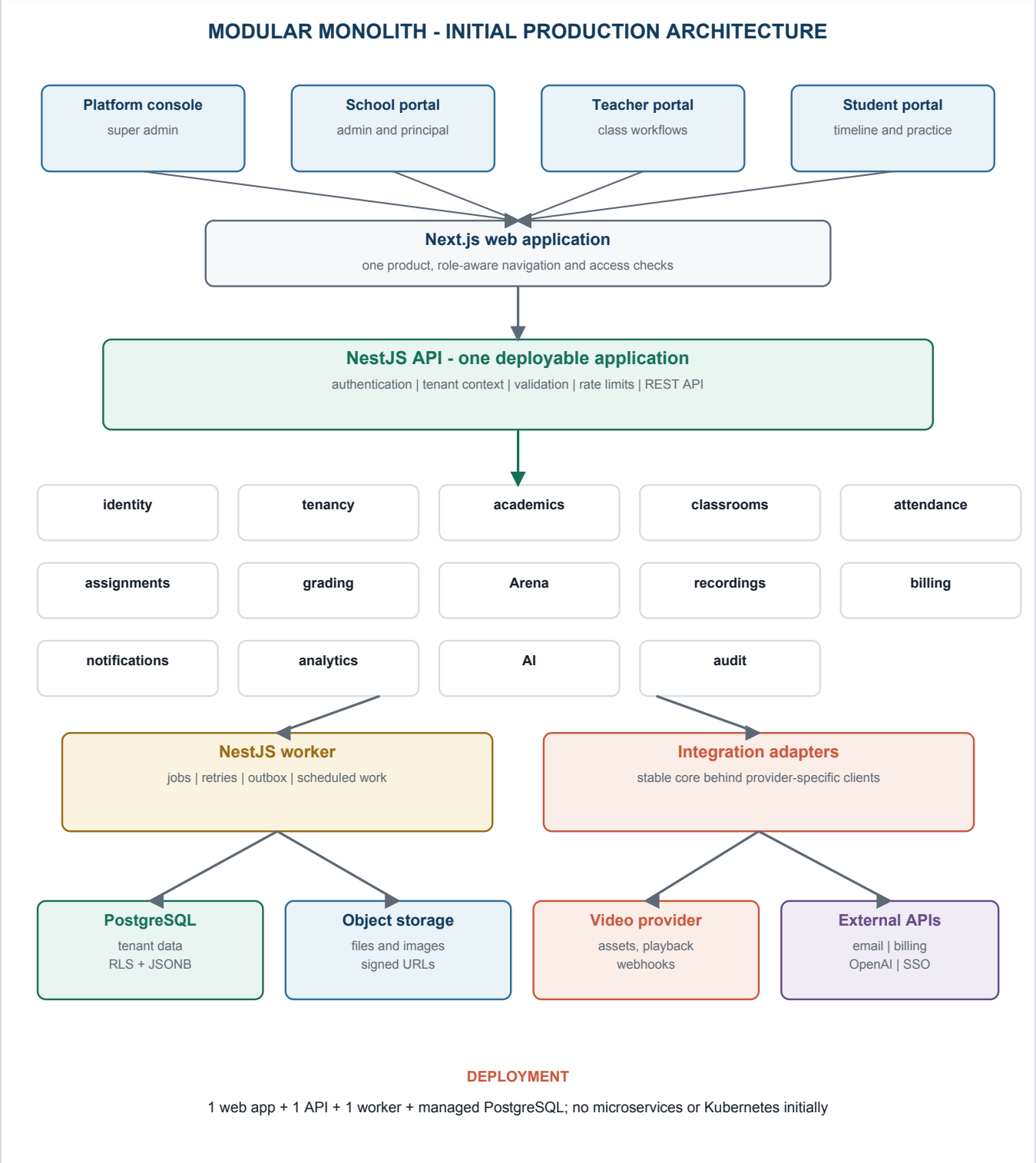


Figure 2. One TypeScript codebase provides explicit module boundaries while avoiding network calls, distributed transactions and independent deployments. The worker uses the same domain modules as the API.

08 | NESTJS AND POSTGRESQL

Backend implementation blueprint

Layer	Initial choice	Upgrade trigger
Frontend	One Next.js app with role-aware route groups	Separate apps only for independently owned experiences
API	NestJS modular monolith, REST and OpenAPI	Split a module for measured load or security ownership
Database	Managed PostgreSQL, one database	Read replicas / warehouse after report load requires it
Flexible data	PostgreSQL JSONB with versioned schemas	Another datastore only for a proven workload
Jobs	pg-boss or transactional outbox + worker	Redis/BullMQ when throughput or latency justifies it
Files	S3-compatible object storage with signed URLs	CDN and lifecycle tiers as volume grows
Video	Mux or Cloudflare Stream asset references	Provider review based on volume and region
AI	OpenAI Responses API from worker	FastAPI only for Python-specific ML pipelines
Deployment	Web, API and worker on managed containers	Kubernetes only when operations require it

Core relational model

Domain	Core records
Tenant	tenants, users, memberships, tenant_entitlements, tenant_settings
Academic	academic_years, terms, subjects, class_sections, teaching_assignments, enrollments
Learning	class_sessions, timeline_entries, materials, recordings, files
Assessment	assignments, questions, submissions, answers, grades, grade_events
Arena	question_tags, practice_sets, arena_attempts, mastery_snapshots
Operations	notifications, outbox_events, webhook_events, audit_events, usage_events

Users are global identities; membership grants a role inside a tenant. Every tenant-owned row carries a non-null tenant_id. Unique keys and indexes normally begin with tenant_id. PostgreSQL row-level security provides a second isolation boundary beyond application authorization.

09 | ONE PRODUCT, CONTROLLED VARIATION

Dynamic configuration without core forks

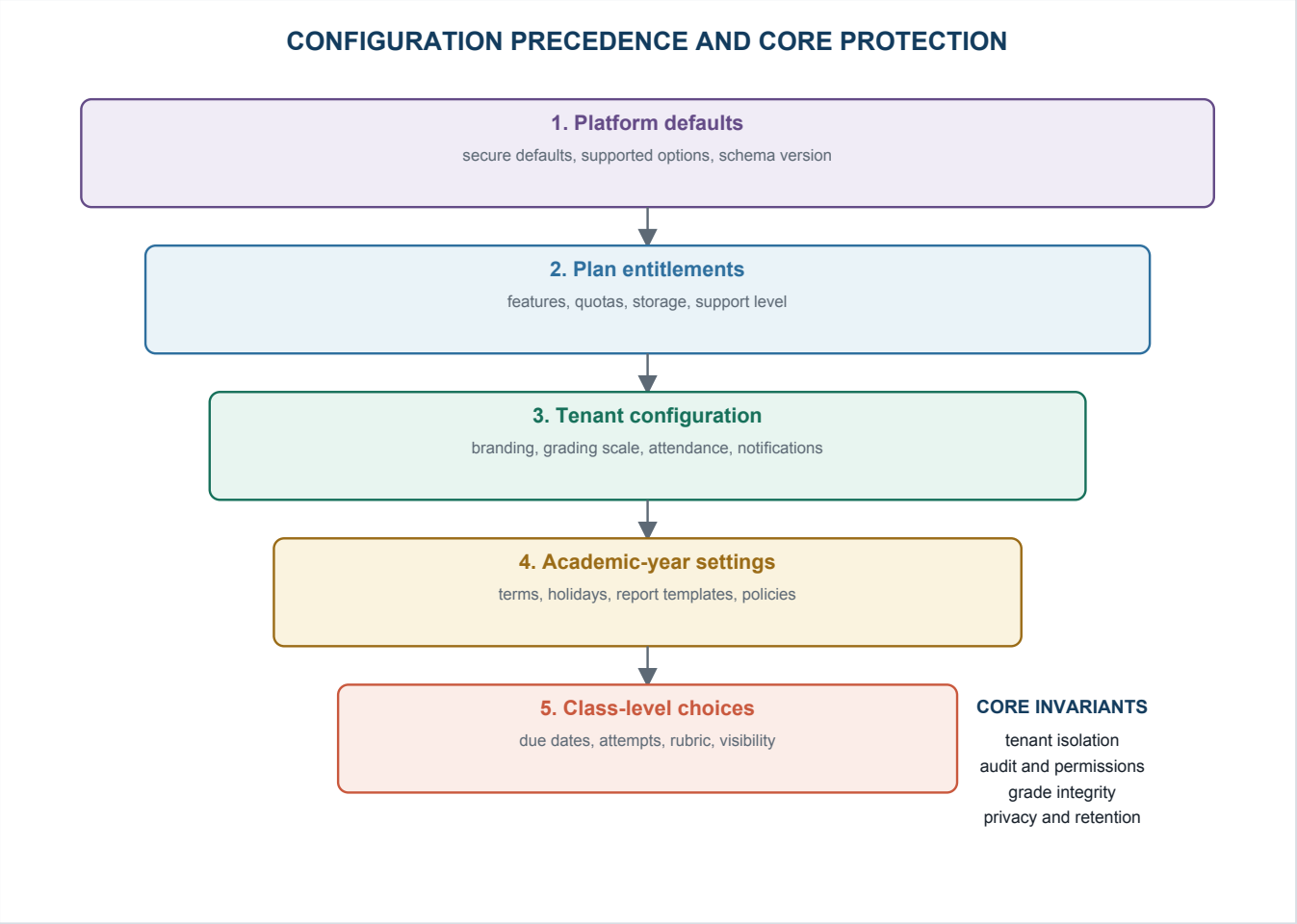


Figure 3. Configuration becomes more specific from platform defaults to class choices. Core security, audit and grade-integrity rules remain invariant and cannot be overridden by tenant configuration.

Configurable area	Implementation pattern	Guardrail
Branding	logo, colors, domain and email identity	No tenant-specific frontend code
Entitlements	feature key, enabled state and numeric quota	Separate commercial entitlement from rollout flags
Academic model	term names, grading scales, attendance labels and holidays	Version by academic year
Workflow options	late policy, attempts, approval and visibility enums	No tenant-authored scripts
Custom fields	versioned field definitions + JSONB values	Supported types, validation and permission scope
Reports	approved templates, column selection and filters	No arbitrary tenant SQL
Notifications	templates, channels and quiet hours	Mandatory security messages cannot be disabled
Integrations	provider adapters, credentials and mappings	Secrets encrypted; webhooks signed and idempotent

10 | PRESERVE PRODUCT INTEGRITY

Configuration implementation rules

Recommended persistence

Record	Purpose
tenant_settings	Versioned JSONB for bounded UI, academic and notification settings
feature_definitions	Platform-owned catalog of supported feature keys and limit types
plan_entitlements	Default features and quotas attached to a commercial plan
tenant_entitlement_overrides	Exceptional enable/disable or quota override with reason and expiry
custom_field_definitions	Tenant-owned labels, types, validation and visibility
integration_connections	Provider, encrypted credential reference, status and field mapping
configuration_audit	Actor, before/after values, timestamp, reason and rollback reference

Rules that prevent customization debt

- Configuration must be represented by supported data, never tenant-specific branches in core code.
- Use a precedence function: platform default -> plan -> tenant -> academic year -> class.
- Validate every settings document against a versioned schema before saving it.
- Keep feature rollout flags separate from customer entitlements.
- Every configuration change is audited and important changes can be rolled back.
- A request from one school becomes a product option only when it is reusable and strategically aligned.
- Use webhooks and standards for extensions; do not create a public plugin runtime in the early product.

80/20 configuration principle: support common variation in data; refuse arbitrary behavior that makes every tenant a separate software product.

11 | TRUST IS A PRODUCT FEATURE

Security, privacy, standards and AI

Control	Minimum production requirement
Tenant isolation	tenant_id on every owned row, RLS policies, cross-tenant integration tests
Authorization	fixed roles plus resource checks: assigned teacher, enrolled student, permitted admin
Platform support	approved break-glass access, expiry, reason and audit trail
Authentication	managed OIDC provider, MFA for privileged roles, secure recovery
Data lifecycle	purpose, retention, export, correction and deletion procedures
Files	signed URLs, malware scanning, type/size limits and lifecycle deletion
Webhooks/jobs	signatures, idempotency keys, retries, dead-letter handling and tenant context
Audit	role, grade, attendance, configuration, billing and support access events
Reliability	automated backups, restore tests, monitoring and incident response

AI operating policy

- Use AI for drafts: questions, rubrics, explanations, feedback and summaries.
- Require teacher approval before publishing educational content or feedback.
- Never use AI alone for final grades, discipline or high-impact student decisions.
- Remove names, emails and direct identifiers before model requests.
- Use Structured Outputs, store=false where appropriate, hashed safety identifiers and tenant cost caps.
- Log model, prompt version, token use, output and teacher disposition for evaluation.

India's DPDP framework requires a deliberate approach to children's data, consent responsibilities, data minimization, security, retention and breach handling. Obtain qualified legal advice before production. Target WCAG 2.2 AA. Start interoperability with CSV import/export shaped around OneRoster; implement full OneRoster/LTI support when sales demand it.

12 | SUSTAINABLE, TESTABLE PRICING

Commercial model and organization pricing

Public benchmarks vary sharply: low-cost Indian tools advertise single-digit INR per student per month; Fedena publicly lists broad annual plans around USD 999-1,699; MoodleCloud lists lower-cost hosted LMS plans for smaller user counts; large vendors commonly require a sales conversation. Competing only on price would make onboarding and support uneconomic.

Offer	Monthly equivalent	Included scale	One-time onboarding	Purpose
Paid pilot	INR 25,000 / 90 days	Up to 1,000 students	Included	Prove activation and value with 3-5 design partners
Essential	INR 7,500 / month	Up to 500 students	INR 30,000	Core LMS, timeline, attendance, assignments and Arena
Growth	INR 15,000 / month	Up to 1,500 students	INR 60,000	Advanced reports, parent digest and larger quotas
Scale	INR 30,000 / month	Up to 4,000 students	INR 120,000	SSO, priority support, integrations and governance
School group	Custom annual contract	Multi-campus / multi-school	Scoped	Group controls, migration, SLA and implementation

Pricing rules

- Bill annually in advance; show a monthly equivalent for comparison.
- Treat these numbers as discovery hypotheses, not final list prices.
- Charge separately for data migration, unusual integrations and custom training.
- Include a sensible AI/video/message allowance; meter expensive overage rather than promising unlimited usage.
- Avoid discounts above 15% without a multi-year commitment or strategic reference value.
- Target more than 75% gross margin after onboarding normalizes.

Price against teacher time saved and school visibility gained - not against raw hosting cost.

13 | DISTRIBUTION AND ADOPTION

How to start and beat competitors

Before writing production code

Action	Required evidence
Interview 20-30 stakeholders	Repeated pain stated independently by owners, admins and teachers
Observe real workflows	Screenshots/forms of attendance, assignments, grading and communication
Choose one segment	One region, board profile, school size and buyer
Prototype the core loop	Teachers complete key tasks without explanation
Secure paid intent	At least three paid pilots or meaningful signed commitments
Define the data agreement	Clear ownership, support access, retention, export and deletion
Set pilot metrics	Activation, weekly teacher use, completion and renewal criteria

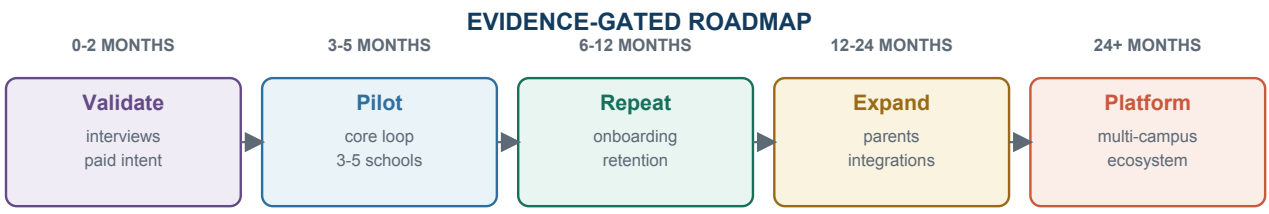
Competitive playbook

Advantage	How to execute
Seven-day go-live	Opinionated setup wizard, CSV import, defaults and assisted migration
Teacher-first usability	Optimize attendance, publish, grade and reuse flows; measure task time
Local service	Founder-led onboarding, training, office hours and fast support
Visible proof	Weekly school adoption and learning-loop report, not vanity dashboards
Trust	Tenant isolation, exports, audit, honest AI policy and transparent incident handling
Integration	Fit existing Google/Microsoft and SIS workflows instead of demanding replacement

Use founder-led sales for the first 10-20 schools. Support them manually enough to understand the work, but convert repeated manual steps into product onboarding. Build channel partnerships only after the sales and implementation motion is repeatable.

14 | EXPAND ONLY THROUGH EVIDENCE

Execution roadmap and future plan



Gate: customer evidence must unlock the next investment

Do not scale engineering ahead of activation, retention, and repeatable sales.

Split services only for measured load, security isolation, or independent ownership.

Phase	Product	Business gate
0. Validate	Prototype, data model, security model and workflow maps	3+ paid pilots or strong signed commitments
1. Pilot	Tenant, roster, classes, attendance, assignments, gradebook, timeline	School live in 7 days; weekly teacher activity
2. Repeat	Arena, notifications, recordings, reports and onboarding automation	Repeatable sales, support load controlled, first renewals
3. Expand	Parent experience, SSO, OneRoster/LTI, advanced analytics	Demand from retained customers and larger contracts
4. Platform	Multi-campus, ecosystem, benchmarking and regional expansion	Strong retention, governance and operational maturity

Company-building sequence

- Wedge: become excellent at the daily classroom learning loop.
- Retention: prove teachers continue using it and schools renew.
- Expansion: add parent, reporting and integration products to existing customers.
- Distribution: create references, implementation partners and board/region playbooks.
- Platform: enable content and integration ecosystems after the core product has pull.
- Geographic growth: localize compliance, curriculum, language and support one market at a time.

15 | WHAT CAN STOP THE COMPANY

Challenges, blockers and risk controls

Risk	Early signal	Control
Scope explosion	Roadmap driven by isolated school requests	ICP, product principles and reusable-configuration test
Teacher non-adoption	Admin uses reports but classroom data stays empty	Teacher task-time targets, training and weekly activation review
Data migration failure	Onboarding takes weeks and rosters are wrong	Validated CSV templates, dry runs and reconciliation report
Tenant data leak	Missing tenant filters or support access shortcuts	RLS, resource authorization tests, audit and incident plan
Support economics	Each school needs unique manual operations	Onboarding fee, standard configuration and productized playbooks
Low willingness to pay	Interest without budget or procurement action	Paid pilots and value-based buyer interviews before scale
School sales seasonality	Deals stall around exams/budget cycles	Region calendar, pipeline planning and annual contracts
Connectivity/device limits	Uploads fail and teachers avoid mobile workflows	Responsive web, resumable uploads, compression and low-bandwidth QA
AI trust/cost	Low teacher acceptance or high usage cost	Human review, evaluation set, quotas and usage reporting
Reporting complexity	Every school requests a different report card	Start with progress views; template only repeated formats

Metrics that decide investment

Category	Measure	Initial target hypothesis
Activation	Contract to first active classes	7 days or less
Teacher adoption	Invited teachers active weekly	70%+ during pilot
Student adoption	Enrolled students active weekly	60%+ where product is launched
Learning loop	Active classes completing the weekly loop	Increase every pilot month
Operations	Support hours per school per month	Declining after onboarding
Commercial	Annual logo retention	90%+ after product-market fit
Economics	Gross margin	75%+ after onboarding normalization

16 | CONCRETE STARTING PLAN

First 90 days

Period	Product work	Customer/business work	Exit gate
Weeks 1-2	Workflow maps, permission matrix, data model draft	Recruit 5 design partners; 10 interviews	One repeated high-value workflow
Weeks 3-4	Clickable prototype and CSV templates	Observe teacher tasks; test pricing	3 schools commit time/data/budget
Weeks 5-8	Tenant/auth, academics, import, attendance and assignment slice	Data agreements; onboarding playbook	One school uses end-to-end flow
Weeks 9-12	Submission, grading, timeline, reports, audit and backups	Launch 3-5 paid pilots; weekly reviews	Activation and usage meet pilot thresholds

Small initial team

Responsibility	Minimum ownership
Founder / product / sales	Customer discovery, pricing, roadmap, pilots and fundraising
Engineering	Two strong full-stack/backend-capable engineers; one owns platform quality
Product design	Part-time or contract initially; teacher workflow usability is critical
Customer success	Founder-led initially, then first dedicated hire when pilots become repeatable
Legal/security	Specialist review for contracts, children's data and security practices

Decision rule: do not build a feature until it supports the target learning loop, closes a real sale, protects trust, or measurably improves retention.

Immediate founder decisions

- Choose the first geography, school board profile and student-count range.
- Choose the first buyer and budget owner: principal, school owner or academic head.
- Choose whether the wedge is homework/assessment, attendance-to-progress, or recorded-class continuity.
- Set the paid-pilot price and success conditions before promising the roadmap.

17 | RESEARCH BASIS

Sources, assumptions and decision notes

Research accessed 6 September 2026. Vendor pages describe vendor claims and public list prices, not independently verified product quality or negotiated enterprise pricing. Market numbers are used for directional planning; obtain updated local data before fundraising or formal forecasts.

Topic	Source
India school and enrollment counts	Ministry of Education - UDISE+ 2024-25
Canvas K-12 positioning	Instructure - Canvas for K-12
Moodle multi-tenancy	MoodleDocs - Multi-tenancy
PowerSchool India product scope	PowerSchool - India solutions
Teachmint integrated platform scope	Teachmint - Integrated School Platform
Fedena features and public pricing	Fedena - Pricing and plans
MoodleCloud public pricing	MoodleCloud - Standard plans
Low-cost India pricing benchmark	School Ease - Public pricing
Roster and grade exchange	1EdTech - OneRoster 1.2
Web accessibility	W3C - WCAG 2.2
India data protection overview	Government of India - DPDP overview
PostgreSQL row security	PostgreSQL documentation - Row security
PostgreSQL JSONB	PostgreSQL documentation - JSON types
OpenAI integration	OpenAI documentation - Responses API

Key assumptions

- India-first private-school entry; regulatory and commercial details change by geography.
- A small experienced product team; delivery dates are planning ranges, not commitments.
- Pricing excludes GST and unusual migration/integration work and assumes annual prepayment.
- The product is learning-operations first, not a complete ERP replacement.
- This document is product and technical strategy, not legal, tax or investment advice.

Final thesis: win a narrow, frequent school workflow; make adoption and trust exceptional; expand into a platform only after retention proves the foundation.